

Problem Statement

BrightTV's subscriber growth has plateaued, and the CEO wants the CVM team armed with data-driven insight before committing to this year's growth objective. The business currently lacks visibility into **which subscriber segments watch most, when consumption peaks and dips, and why** — without that, any acquisition or retention initiative is a guess. The session-level dataset (10,000 viewing records across 4,386 subscribers, Jan–Apr 2016) gives us the raw material to close that gap, but it also reveals a data quality issue worth flagging upfront: **~69% of records are classified "No Usage"** in the Screen_time_bucket field despite having a logged session — this needs to be understood (tuning/zapping events vs. real viewing) before KPIs are trusted at face value.

Business Objectives

1. **Grow the subscription base** for the current financial year (the CEO's core mandate).
2. **Understand viewership behavior** by segment — sex, ethnicity, age group, and region are all captured, so we can see who engages most and who's underrepresented.
3. **Reduce low-consumption periods** by identifying which days/times/channels see the weakest engagement and matching content to fix them.
4. **Equip the CVM team** with initiatives (not just stats) that plug into acquisition, retention, and reactivation.

Project Overview

Scope of the analysis, based on the actual fields available:

- **Timezone:** RecordDate_SAST is already SA-time converted; Watch_time/Hour_of_day support day-part analysis directly.
- **Usage trends:** analyze by Day_classification (Weekday/Weekend), Time_of_day (Midnight/Morning/Afternoon/Evening), and Screen_time_bucket (No/Low/Medium/High usage).
- **Segmentation:** cross-cut by Sex, Ethnicity, Age_group, and Regions to find which factors correlate with higher/lower consumption.
- **Content:** Tv_channel lets us rank which channels/genres drive the most engaged (Medium/High usage) sessions vs. which correlate with "No Usage."
- **Contactability:** Email_flag / Social_media_handle_flag indicate which subscribers are reachable for CVM campaigns — relevant for initiative targeting.
- **Output:** 20-minute presentation to the CEO/CVM stakeholders covering trends, drivers, content recommendations, and growth initiatives.

Success Metrics (KPIs)

KPI	Source field(s)	Why it matters
Net new subscribers / growth rate	External subscription data (not in this file)	Direct measure of the CEO's objective
Session volume by day/time-of-day	Day_classification, Time_of_day, Hour_of_day	Pinpoints low-consumption windows
% sessions in Medium/High usage vs. No/Low usage	Screen_time_bucket, Duration_hours	True engagement, not just tuning-in
Engagement by segment	Sex, Ethnicity, Age_group, Regions	Flags under-served or high-value segments
Top/bottom performing channels	Tv_channel	Informs content recommendations for weak days
Reachable subscriber base	Email_flag, Social_media_handle_flag	Feasible audience for CVM campaigns
Uplift in consumption post-initiative	Future tracking	Validates whether recommendations worked

Note: the file has no explicit subscription/churn status field — if BrightTV wants churn/retention as a KPI, that data would need to be joined in separately.

Data Sources Description

Single dataset — "Bright_Tv_Data_set.xlsx", session-level, one row per viewing session (10,000 rows, 4,386 unique subscribers, Jan 1 – Apr 1, 2016):

Field group	Columns
Subscriber identity & demographics	Sub_ID, Sex (male/female/unknown), Ethnicity (black/coloured/indian_asian/white/unknown), Age_group (Kids, youth, youth Adults, Adults, Elder, Pensioner, infant), Regions (9 SA provinces + Unclassified)
Contactability flags	Email_flag, Social_media_handle_flag
Session timing	RecordDate_SAST (converted timestamp), Watch_date, Day_name, Month_name, Event_year, Event_day, Hour_of_day, Day_classification (Weekday/Weekend), Watch_time (fraction of day), Time_of_day (Midnight/Morning/Afternoon/Evening)
Session duration & engagement	Duration, Duration_hours, Duration_seconds, Screen_time_bucket (No/Low/Medium/High usage)

Field group	Columns
Content	Tv_channel

A quick profile: the base skews heavily **male (88%)** and **Gauteng-based (37%)**, with **"youth Adults" the dominant age segment (60%)**. Evening is the busiest time-of-day (40% of sessions), and weekdays outnumber weekends roughly 7:3.

Want me to move on to the actual analysis (charts on usage by day/time/segment, top content by engagement) and start building toward the 20-minute presentation, or would you like this framing turned into a Word doc first?